

All-Genus 2D $\mathcal{N} = 1$ Superconformal Blocks

Sam Christian,^{*} Yuchen Wang,[†] Xi Yin,[‡] and Yutai Zhang[§]
Jefferson Physical Laboratory, Harvard University, Cambridge, MA 02138 USA
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We develop algorithms for numerically computing two-dimensional $\mathcal{N} = 1$ superconformal blocks at arbitrary genera, in arbitrary plumbing channels, and for arbitrary spin structures. For the blocks whose intermediate supermultiplets are all in the Neveu-Schwarz sector, we derive a recursion relation in the central charge at all genera. For the blocks with Ramond intermediate supermultiplets, we utilize a Virasoro $\oplus$ Virasoro subalgebra of the tensor product of the $\mathcal{N} = 1$ superconformal algebra with a free Majorana fermion algebra, which allows us to reduce the superconformal blocks to Virasoro conformal blocks.

INTRODUCTION

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Superconformal blocks are central objects

c -RECURSION FOR NS BLOCKS

^{*} samchristian@fas.harvard.edu

[†] yuchen_wang@fas.harvard.edu

[‡] xiyin@fas.harvard.edu

[§] yutaizhang@fas.harvard.edu

SUPERCONFORMAL BLOCKS FROM DOUBLE VIRASORO

DISCUSSION

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